GATE ME 90 Days Challenge

Day 34/90 · Practice Set with Detailed Solutions

SUBJECT	TODAY'S TOPIC	COUNTDOWN
Fluid Mechanics	**Turbulent Flow & Pipe Losses**	**72 days to GATE**

How to use this set: 15 questions (8 MCQ, 7 NAT) on turbulent pipe flow (Blasius and Moody chart, velocity profiles, viscous sublayer, mixing length), Darcy–Weisbach losses, minor losses at enlargements, contractions, exits and fittings, pipes in series and parallel, siphons and power transmission. Try them in **45 minutes** without looking at the solutions, then check the answer key and the step-by-step solutions. For NAT questions, enter the number; there is no negative marking for NAT in GATE.
Today's test: open 7 to 9 PM in the **GATE ME 90 Days Challenge** course on our app and website (enrol once for ₹1: iurlk.courses.store/909124), 20 new questions on the same topic. Top scorers are announced in the next 8 AM challenge reel.

Questions

Q1 NAT

Water (kinematic viscosity 10^{-6} m²/s) flows at 2 m/s through a smooth 50 mm diameter pipe. Using the Blasius correlation $f = 0.316/Re^{0.25}$ and g = 9.81 m/s², the head loss per 100 m of pipe (in m, round off to two decimal places) is _____

Q2 MCQ

Which statement about the Moody chart is **NOT** correct?

(A) In laminar flow f = 64/Re for all pipes, rough or smooth

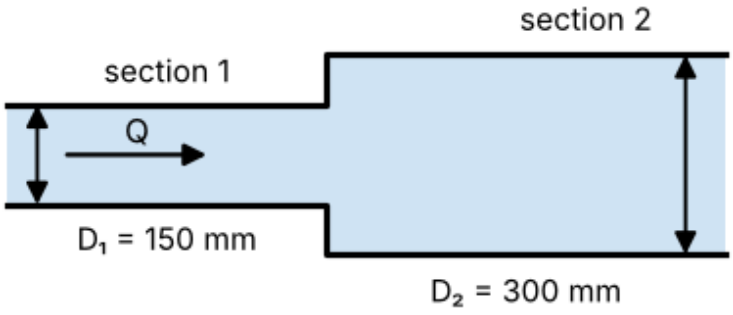
(B) In the transition zone f depends on both Re and ε/D (Colebrook equation)

(C) In the fully rough zone f increases steadily with Re

(D) For a given Re in turbulent flow, a rougher pipe has a larger f

Q3 NAT

A horizontal pipe carrying 0.1 m³/s of water suddenly enlarges from 150 mm to 300 mm diameter, as shown. Take g = 9.81 m/s². The head loss at the enlargement (in m, round off to two decimal places) is _____



Q4 MCQ

For the sudden enlargement in Q3, the rise in pressure head from section 1 to section 2 is about

(A) 0.61 m
(B) 1.53 m
(C) 0.92 m
(D) 0 (pressure falls)

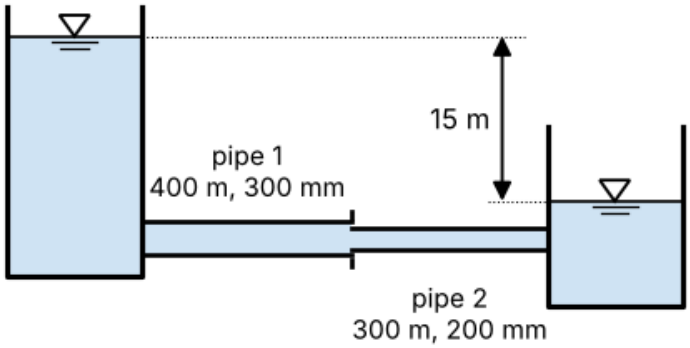
Q5 MCQ

For turbulent pipe flow with the one-seventh power-law profile $u = u_{max}(y/R)^{1/7}$, the momentum correction factor β is about

(A) 1.33
(B) 1.02
(C) 2.00
(D) 1.06

Q6 NAT

Two reservoirs with a 15 m difference in water level are connected by two pipes in series, as shown: 400 m of 300 mm diameter followed by 300 m of 200 mm diameter, both with f = 0.02. Neglecting minor losses (g = 9.81 m/s²), the discharge (in L/s, round off to one decimal place) is _____



Q7 MCQ

Two pipes A and B connect the same two junctions (in parallel). Pipe B has the same length and friction factor as A but twice its diameter. The ratio of discharges Q_B/Q_A is about

(A) 2
(B) 4
(C) 5.66
(D) 32

Q8 NAT

Water (density 1000 kg/m³, kinematic viscosity 10^{-6} m²/s) flows at 2 m/s in a pipe with Darcy friction factor 0.02. Using $\delta' = 11.6\nu/u_*$, the thickness of the viscous sublayer (in mm, round off to three decimal places) is _____

Q9 NAT

A pipe 1500 m long and 150 mm in diameter (Darcy f = 0.02) ends in a nozzle. Neglecting losses other than pipe friction, the nozzle diameter that gives maximum power in the jet (in mm, round off to one decimal place) is _____

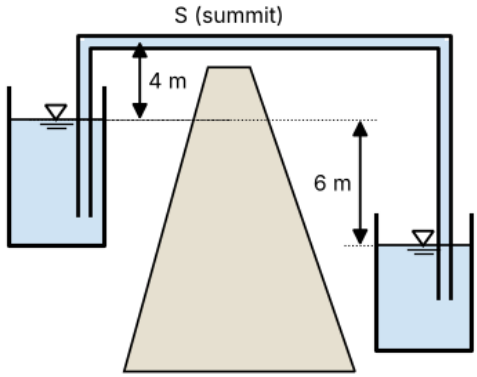
Q10 MCQ

Water flows from a pipe into a large reservoir with velocity V. The head lost at the outlet is

(A) $0.5V^2/2g$
(B) $V^2/2g$
(C) zero, since the reservoir is large
(D) $2V^2/2g$

Q11 NAT

A 100 mm diameter siphon, 300 m long (f = 0.02), connects two reservoirs whose water levels differ by 6 m, as shown. Its summit S is 4 m above the upper water level and 100 m (of pipe length) from the inlet. Neglecting minor losses (g = 9.81 m/s²), the gauge pressure head at S (in m of water, round off to one decimal place) is _____



Q12 MCQ

Near a smooth wall in turbulent flow, Prandtl's mixing length l is taken proportional to the distance y from the wall ($l = \kappa y$, $\kappa \approx 0.4$). Combined with $\tau \approx \tau_w$ near the wall, this gives a velocity profile that is

(A) parabolic in y
(B) linear in y
(C) logarithmic in y
(D) a 1/7 power of y exactly

Q13 MCQ

A 90° elbow has a loss coefficient K = 0.9. In a 100 mm diameter pipe with Darcy f = 0.02, its equivalent length of straight pipe is

(A) 4.5 m
(B) 0.45 m
(C) 45 m
(D) 2.25 m

Q14 MCQ

In a long pipeline operating in the fully rough turbulent regime, the discharge is increased by 20 %. The frictional head loss increases by about

(A) 20 %
(B) 32 %
(C) 44 %
(D) 73 %

Q15 NAT

A pipeline 1000 m long must carry 0.06 m³/s of water with a frictional head loss of 10 m. Taking Darcy f = 0.02 and g = 9.81 m/s², the required pipe diameter (in mm, round off to the nearest integer) is

_____

Answer Key

Q	Type	Answer
Q1	NAT	7.25 (accept 7.18 to 7.32)
Q2	MCQ	(C) In the fully rough zone f increases steadily with Re
Q3	NAT	0.92 (accept 0.91 to 0.93)
Q4	MCQ	(A) 0.61 m
Q5	MCQ	(B) 1.02
Q6	NAT	90.8 (accept 90.0 to 91.5)
Q7	MCQ	(C) 5.66
Q8	NAT	0.116
Q9	NAT	33.5 (accept 33.2 to 33.9)
Q10	MCQ	(B) $V^2/2g$
Q11	NAT	-6.1
Q12	MCQ	(C) logarithmic in y
Q13	MCQ	(A) 4.5 m
Q14	MCQ	(C) 44 %
Q15	NAT	226 (accept 225 to 228)

Detailed Solutions

Q1 **Answer: 7.25 (accept 7.18 to 7.32)**

Concept: Find Re, then f from Blasius, then use Darcy–Weisbach $h_f = fLV^2/(2gD)$.

$Re = VD/\nu = 2 \times 0.05/10^{-6} = 10^5$ (turbulent, within the Blasius range)

$f = 0.316/(10^5)^{0.25} = 0.01777$

$h_f = 0.01777 \times (100/0.05) \times 2^2/(2 \times 9.81) =$ **7.25 m**

Exam tip: Blasius gives $h_f \propto V^{1.75}$ for smooth pipes; fully rough pipes give $h_f \propto V^2$.

Q2 **Answer: (C) In the fully rough zone f increases steadily with Re**

Concept: The Moody chart has laminar, critical, transitional-turbulent and fully rough zones.

In the fully rough zone the curves are **horizontal**: f depends only on ε/D, not on Re. So (C) is wrong.

Exam tip: Fully rough: $1/\sqrt{f} = 2\log_{10}(3.7D/\varepsilon)$. Smooth: Blasius or Prandtl's $1/\sqrt{f} = 2\log_{10}(Re\sqrt{f}) - 0.8$.

Q3 **Answer: 0.92 (accept 0.91 to 0.93)**

Concept: Borda–Carnot loss at a sudden enlargement: $h_L = (V_1 - V_2)^2/(2g)$.

$V_1 = 0.1/(\pi/4 \times 0.15^2) = 5.6588$ m/s; $V_2 = V_1 \times (150/300)^2 = 1.4147$ m/s

$h_L = (5.6588 - 1.4147)^2/19.62 =$ **0.92 m**

The pressure still **rises** across the enlargement: $(p_2 - p_1)/\rho g = V_2(V_1 - V_2)/g = 0.612$ m.

Exam tip: When the larger pipe is a reservoir ($V_2 = 0$), the same formula gives the exit loss $V_1^2/2g$.

Q4 **Answer: (A) 0.61 m**

Concept: Energy equation: $p_1/\rho g + V_1^2/2g = p_2/\rho g + V_2^2/2g + h_L$.

$(p_2 - p_1)/\rho g = (V_1^2 - V_2^2)/2g - (V_1 - V_2)^2/2g = V_2(V_1 - V_2)/g$

$= 1.4147 \times 4.2441/9.81 =$ **0.61 m**

Exam tip: Without losses the rise would be $(V_1^2 - V_2^2)/2g = 1.53$ m; the loss eats exactly the difference.

Q5 **Answer: (B) 1.02**

Concept: $\beta = (1/A)\int(u/\bar{u})^2\,dA$. For the 1/n power law in a pipe, $\bar{u}/u_{max} = 2n^2/((n+1)(2n+1))$ and $\beta = (n+1)(2n+1)^2/(4n^2(n+2))$.

$n = 7$: $\bar{u}/u_{max} = 98/120 = 0.817$; $\beta = 8 \times 225/(4 \times 49 \times 9) =$ **1.02**

Exam tip: Turbulent profiles are flat, so $\alpha \approx 1.06$ and $\beta \approx 1.02$ are usually taken as 1. Laminar pipe flow: $\alpha = 2$, $\beta = 4/3$.

Q6 **Answer: 90.8 (accept 90.0 to 91.5)**

Concept: In series the same Q flows through both pipes and the friction losses add up to the level difference. Write $h_f = 8fLQ^2/(\pi^2 g D^5)$.

$15 = (8 \times 0.02\,Q^2/(\pi^2 \times 9.81)) \times (400/0.3^5 + 300/0.2^5) = 1821.3\,Q^2$

$Q = \sqrt{(15/1821.3)} = 0.09075$ m³/s = **90.8 L/s**

Exam tip: The 200 mm pipe contributes $300/0.2^5 = 9.4 \times 10^5$ against 1.6×10^5 for the bigger pipe: the smaller pipe controls the flow.

Q7 **Answer: (C) 5.66**

Concept: Parallel pipes have the same head loss: fLQ^2/D^5 is equal $\Longrightarrow Q \propto D^{2.5}$ (same f and L).

$Q_B/Q_A = 2^{2.5} =$ **5.66**

Exam tip: Velocity ratio is $2^{0.5} = 1.41$ ($V \propto \sqrt{D}$ at the same h_f/L); discharge picks up another D^2.

Q8 **Answer: 0.116**

Concept: $\tau_w = f\rho V^2/8$ and friction velocity $u_* = \sqrt{(\tau_w/\rho)}$.

$\tau_w = 0.02 \times 1000 \times 2^2/8 = 10$ Pa; $u_* = \sqrt{(10/1000)} = 0.1$ m/s

$\delta' = 11.6 \times 10^{-6}/0.1 = 1.16 \times 10^{-4}$ m = **0.116 mm**

Exam tip: A roughness of 0.05 mm here gives $k/\delta' = 0.43$, between 0.25 and 6: the pipe is in the transition regime.

Q9 **Answer: 33.5 (accept 33.2 to 33.9)**

Concept: Jet power is maximum when $h_f = H/3$, i.e. jet velocity head $v^2/2g = 2h_f$.

$v^2/2g = 2fLV^2/(2gD)$ and $v = V(D/d)^2 \Longrightarrow (D/d)^4 = 2fL/D \Longrightarrow d = (D^5/(2fL))^{1/4}$

$d = (0.15^5/(2 \times 0.02 \times 1500))^{1/4} = 0.03354$ m = **33.5 mm**

Exam tip: If the question uses the Fanning-type f in $h_f = 4fLV^2/(2gD)$, the formula becomes $d = (D^5/(8fL))^{1/4}$.

Q10 **Answer: (B) $V^2/2g$**

Concept: The jet's kinetic energy is entirely dissipated in the still reservoir: a sudden enlargement with $V_2 = 0$.

$h_{exit} = (V - 0)^2/2g =$ **$V^2/2g$** ($K = 1.0$).

Exam tip: Typical K values: sharp entrance 0.5, rounded entrance 0.04, exit 1.0, 90° elbow about 0.9.

Q11 **Answer: −6.1**

Concept: All of the 6 m difference is spent in friction over 300 m; then apply energy from the upper free surface to S.

$6 = (fL/D)(V^2/2g) = 60 \times V^2/2g \Longrightarrow V^2/2g = 0.1$ m ($V = 1.401$ m/s)

Friction from inlet to S: $h_{f1} = 0.02 \times (100/0.1) \times 0.1 = 2.0$ m

$0 = p_S/\rho g + V^2/2g + 4 + h_{f1} \Longrightarrow p_S/\rho g = -(4 + 0.1 + 2.0) =$ **-6.1 m**

Absolute: $10.3 - 6.1 = 4.2$ m of water, safely above the vapour-pressure limit.

Exam tip: The pressure is lowest at the summit, not at the highest velocity point, because elevation dominates.

Q12 **Answer: (C) logarithmic in y**

Concept: $\tau_w = \rho\kappa^2y^2(du/dy)^2 \Longrightarrow du/dy = u_*/(\kappa y)$.

Integrating: $u/u_* = (1/\kappa) \ln y + C$, i.e. the **logarithmic** law ($u/u_* = 5.75 \log_{10}(yu_*/\nu) + 5.5$ for smooth pipes).

Exam tip: Inside the viscous sublayer the profile is linear: $u/u_* = yu_*/\nu$.

Q13 **Answer: (A) 4.5 m**

Concept: Equivalent length: $K\,V^2/2g = f(L_e/D)V^2/2g \Longrightarrow L_e = KD/f$.

$L_e = 0.9 \times 0.1/0.02 =$ **4.5 m**

Exam tip: Adding equivalent lengths of all fittings to the pipe length lets you do the whole system with one Darcy–Weisbach term.

Q14 **Answer: (C) 44 %**

Concept: Fully rough: f is constant, so $h_f \propto Q^2$.

$h_{f2}/h_{f1} = 1.2^2 = 1.44 \Longrightarrow$ increase ≈ **44 %**

Exam tip: Pumping power $\propto Qh_f \propto Q^3$, so the power rises by $1.2^3 - 1 = 73$ %.

Q15 **Answer: 226 (accept 225 to 228)**

Concept: In terms of discharge, $h_f = 8fLQ^2/(\pi^2 g D^5)$; solve for D.

$D^5 = 8 \times 0.02 \times 1000 \times 0.06^2/(\pi^2 \times 9.81 \times 10) = 5.9491\text{e-}04\ m^5$

D = **226 mm** (choose the next standard size, e.g. 250 mm).

Exam tip: $D \propto Q^{0.4}$: doubling the flow at the same head loss needs only $2^{0.4} = 1.32$ times the diameter.

What next?

Day 34 test tonight, 7 to 9 PM

20 questions · 60 minutes · GATE marking (1/3 and 2/3 negative). Top 3 scorers are announced with name, marks and photo (if shared) in the next 8 AM challenge reel. Not enrolled yet? Join the ₹1 challenge course: iurlk.courses.store/909124

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